To fully leverage the power of gen AI, robust and scalable infrastructure is paramount. This is where Google Kubernetes Engine becomes a critical component. GKE provides a managed environment for deploying, managing, and scaling containerized applications, making it ideal for the demanding workloads of gen AI. This module will explore how GKE facilitates the deployment and optimization of gen AI models, covering topics such as resource management, scaling strategies, and integration with Google Cloud's AI/ML services.

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GKE and AI

01 AI and GKE overview

02 AI model training on GKE

03 AI model serving on GKE

04 AI cost management on GKE

Google Cloud

In this section, titled “GKE and AI”, you learn to:

- Explain how GKE serves as a suitable platform for large language models and the increasing demand for hardware accelerators.
- Describe the high-level architecture of a GKE-based training platform for AI models.
- Outline the architecture for a GKE-based model serving platform.
- Outline different cost management strategies available when using GKE for AI/ML workloads.

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GKE and AI

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Google Cloud

Let’s start by exploring the impact of gen AI as it relates to GKE at a high-level.

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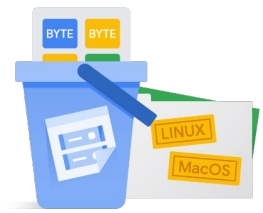
The approach to LLMs is constantly evolving

Hyperscaling model sizes	Growing OSS ecosystem	Increased demand for accelerators
		

Google Cloud

The ways in which organizations approach large language models is constantly changing.

Model sizes are growing rapidly, resulting in a fundamental shift in model capabilities.

The OSS software ecosystem is growing too. Organizations want to make the most of ML with an open software ecosystem, providing easy access to open-source tools, open models, frameworks, and APIs.

And the demand for hardware accelerators for AI/ML workloads has increased substantially.

To meet the demands of this hyperscaling trend, organizations require a platform that can scale dynamically to very high limits. Large-scale model training and inference are expensive and ongoing, emphasizing the need for cost-efficient scaling methods.

Additionally, a single platform that supports all workload types, including AI workloads, simplifies the management and optimization of an organization's infrastructure.

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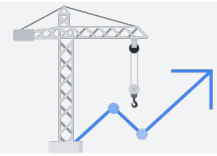
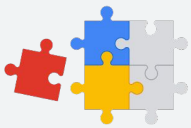
GKE supports any workload type, including AI

Consistent and robust experience

Unmatched scalability



Flexible accelerator options



Rich ecosystem of 1st and 3rd party frameworks

Google Cloud

Kubernetes is an excellent platform for AI workloads. It supports any workload type including AI, driving robust and consistent development across the stack.

GKE provides a single platform for all your workloads, ensuring a consistent and robust development process.

As the foundation, it's scalable and compatible with a diverse set of hardware accelerators including GPUs and Google proprietary TPUs. Accelerator orchestration helps improve performance and reduce costs.

Google's open model for Gen AI, Gemma, is available on GKE. Gemma is a family of lightweight, state-of-the-art open models built from the same research and technology that's used for Gemini models. Gemma on GKE empowers teams to deploy and easily serve open models.

Kubernetes (and GKE by extension) has access to an ecosystem of first-party and third-party frameworks such as TensorFlow, Ray, and PyTorch, making it easy to build or extend AI models.

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Workload optimized infrastructure

AI and ML workloads

Machine Learning
Deep learning
Recommender systems
Natural language
Computer vision

Modern workloads

Containerized
Serverless
Web3
Streaming
Data analytics

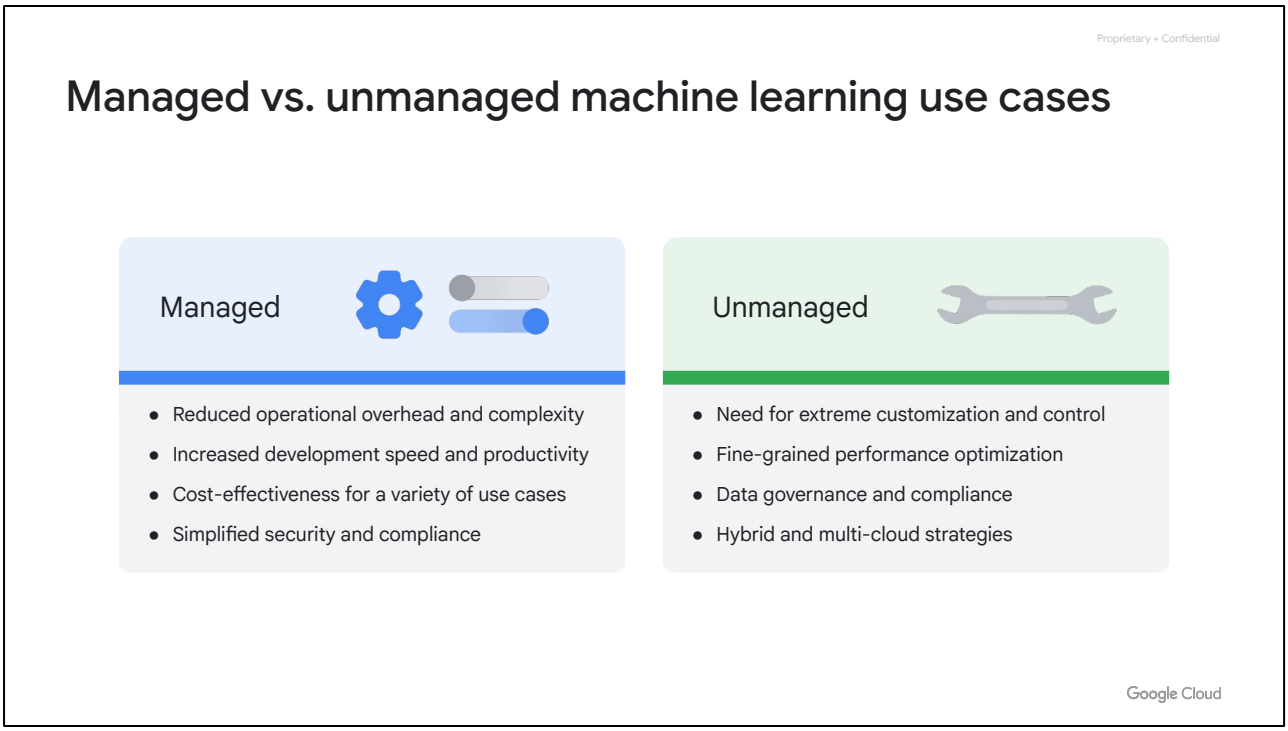
Traditional enterprise workloads

SAP
VMware
Microsoft
HPC
Custom 3-tier

Google Cloud

To optimize your infrastructure, assess each workload. Then design and optimize at a system level across compute, storage, networking, hardware and software, to meet the unique needs of each workload.

Factors to consider include performance, reliability, security, cost, and sustainability.



Google Cloud offers a wide range of managed services that integrate with Vertex AI and ML workloads.

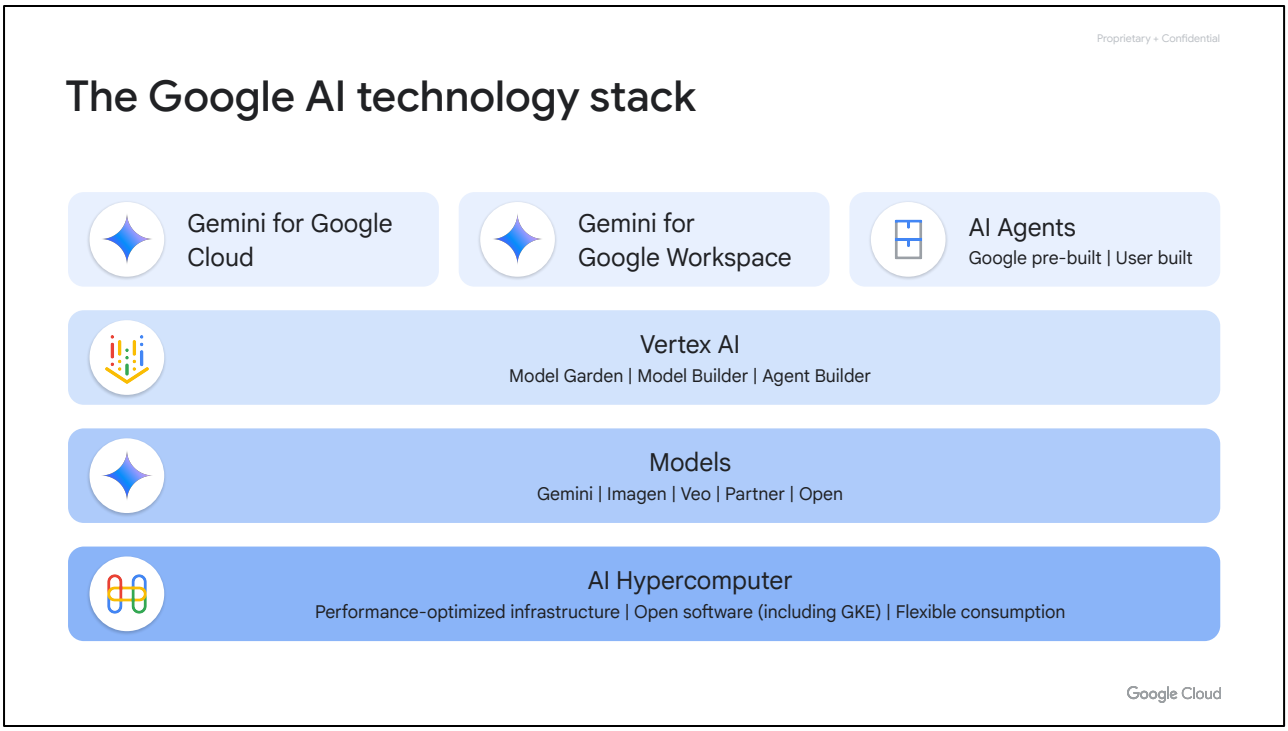
Let's explore the benefits of using these managed services.

- **Reduced operational overhead and complexity:** Assisting with elements like infrastructure management, scalability, availability, and maintenance.
- **Increased development speed and productivity:** Providing simplified ML workflows, pre-built components, and tools.
- **Cost-effectiveness for a variety of use cases:** Pay-as-you-go pricing and reduced operational costs contribute to optimized resource utilization.
- **Simplified security and compliance:** Google provides built-in security and compliance certifications.

But managed services are not always the most suitable approach. Use cases for unmanaged services include:

- **Need for extreme customization and control:** Using custom algorithms and models, requiring unique pre-processing and post-processing, hardware, and software dependencies.
- **Fine-grained performance optimization:** For applications requiring extremely

- low-latency inference, specific resource management and cost optimization requirements.
- **Data governance and compliance:** Some industries or regions have strict regulations with data residency and sovereignty. Sensitive data may require customized security and access control. Auditability and explainability may not be available with managed services.
- **Hybrid and multi-cloud strategies:** Existing infrastructure may have large investments in existing infrastructure.



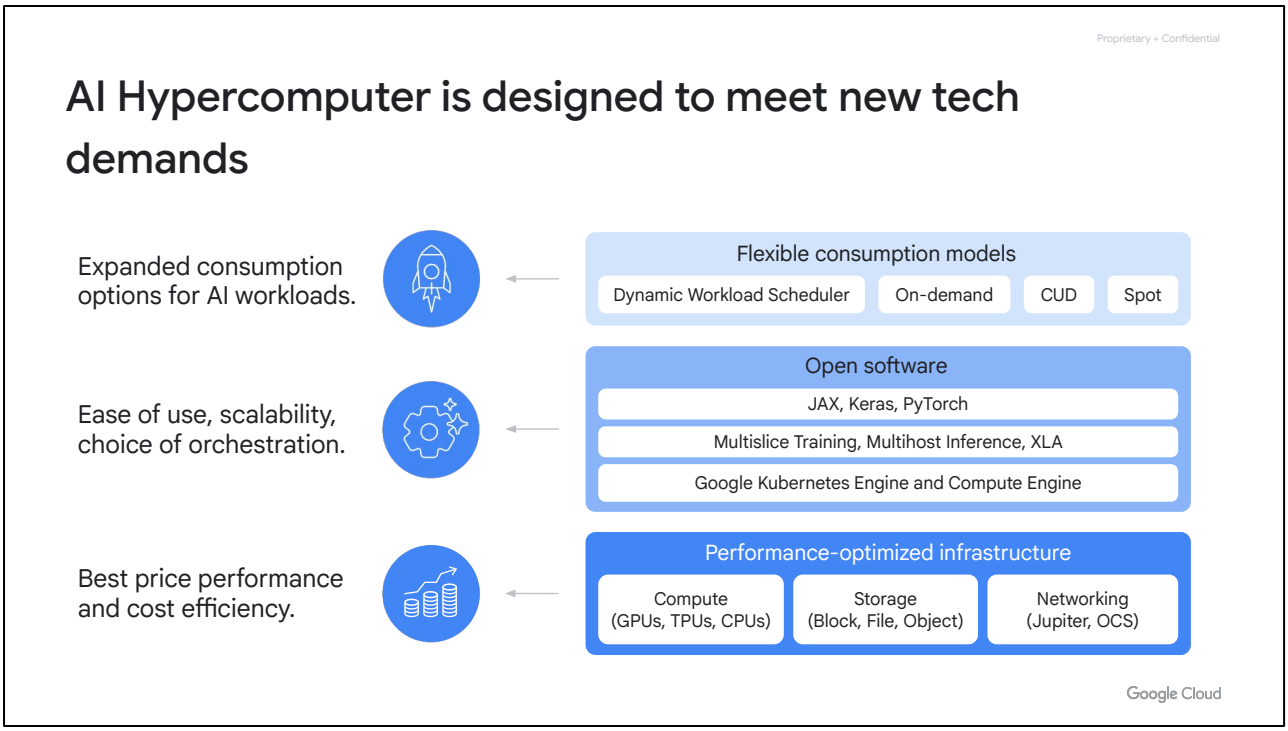
The Google technology stack is fully integrated and open, enabling organizations to bring gen AI to real-world experiences quickly, efficiently, and responsibly.

Let's break down the components of this technology stack.

At the top of the infrastructure is **Vertex AI**, a platform that provides the tools and infrastructure to build, deploy, and manage your agents.

Then, there are **Gemini models**, the brains behind the agents.

And finally, the foundation is the **AI Hypercomputer**, which provides the power and scalability needed to train and serve AI models.



Let's explore the architecture of the AI Hypercomputer.

At the foundation is a selection of purpose-built infrastructure offerings. For compute options there are Cloud TPUs, GPUs, and CPUs. For storage, options like GCS Fuse, Parallelstore, and Hyperdisk ML are available to address varying storage needs. And you have the core data center expertise that's used to serve AI experiences to billions of Google users globally.

Because GPUs and GPU resources must be portable, containerization is an ideal solution.

GKE can handle orchestration across thousands of GPUs.

GPUs are in high demand, and therefore expensive. With GKE, you get maximum performance for the cost.

Next is software. APIs and other primitives help command the hardware, and GKE and partner offerings help schedule and orchestrate workloads.

The next level consists of the frameworks, libraries, and tools where ML practitioners need the most flexibility. These elements may come from different projects.

Flexible consumption options let you experiment while still being cost-efficient. Latency tolerants or fixed duration workloads enable experimentation without

runaway costs.

When you put everything together, you get a collection of patterns and the confidence in knowing they will perform well over time without sacrificing flexibility.

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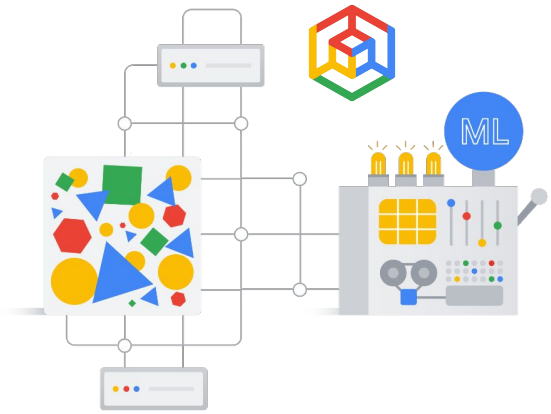
Benefits of using GKE for model training



- Portability
- Efficient orchestration
- Maximum performance
- Cost effective

Google Cloud

Portability, efficient orchestration, and cost effective, maximum performance are benefits of using GKE for AI model training.

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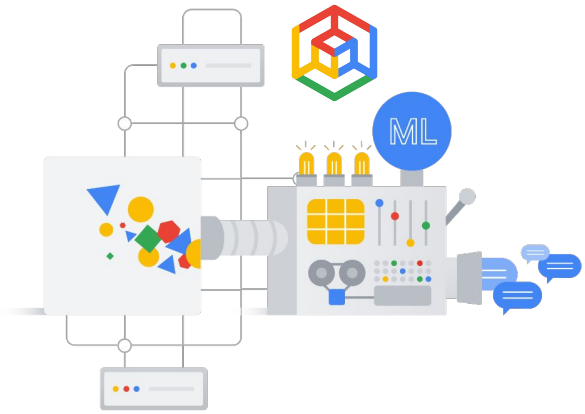
Benefits of using GKE for model serving



- Lower inference costs
- Increased AI engineering velocity
- Fine-grained control
- Cost effective

Google Cloud

In addition to model training, there are several benefits of using GKE for model serving. Open models are driving down inference costs, and GKE tools and best practices increase AI engineering velocity. To scale and optimize your workloads, fine-grained control is necessary.

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Google Cloud

Training AI models is a complex and time consuming process, let's explore how GKE can assist the process.

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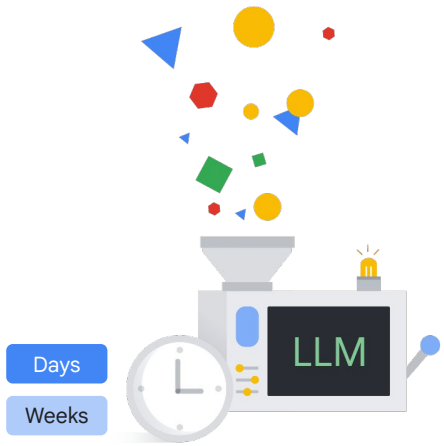
Challenge: model training at scale



- Large datasets and complex models requiring substantial compute power.
- Extended training time on standard infrastructure.
- Resource utilization and scaling training jobs is paramount.
- Often demand custom environments and specific dependencies.

Google Cloud

AI model training often involves massive datasets and complex models requiring substantial compute power.

Training can last for days or even weeks on standard infrastructure, hindering development velocity.

Efficient resource utilization and the ability to scale training jobs are paramount.

Furthermore, many advanced training scenarios demand custom environments and specific dependencies that managed services may not readily provide.

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Use case: distributed image classification

Google Cloud

Let's consider a practical use case.

Imagine you're training a deep learning model to categorize millions of images. To expedite the training process, you use distributed training techniques.

This example involves custom image pre-processing and data augmentation to enhance model performance. Training framework flexibility is also important to enable the use of Tensorflow or PyTorch as needed.

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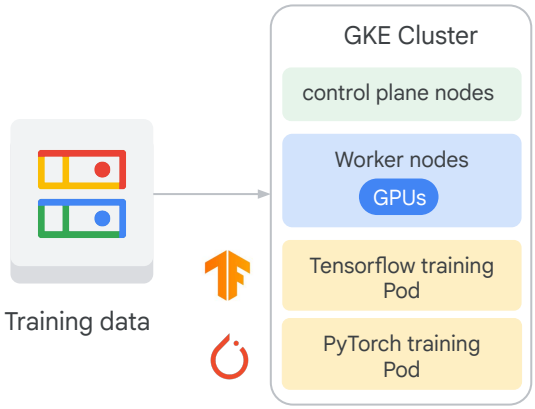
GKE-based training platform



Google Cloud

So what might the high-level architecture of a GKE-based training platform be?

The training data is stored in Cloud Storage. You can use a GKE cluster with control plane and worker nodes. Some worker nodes are equipped with GPUs for accelerated training.

Within the GKE cluster, deploy training pods that run distributed TensorFlow or PyTorch jobs.

Optionally, you can leverage the model registry and deployment capabilities of Vertex AI to leverage its model registry and deployment capabilities of Vertex AI.

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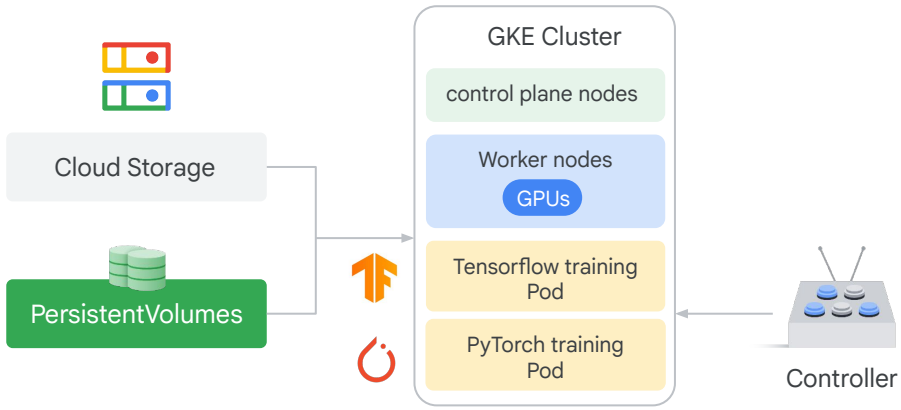
GKE-based training platform setup



Google Cloud

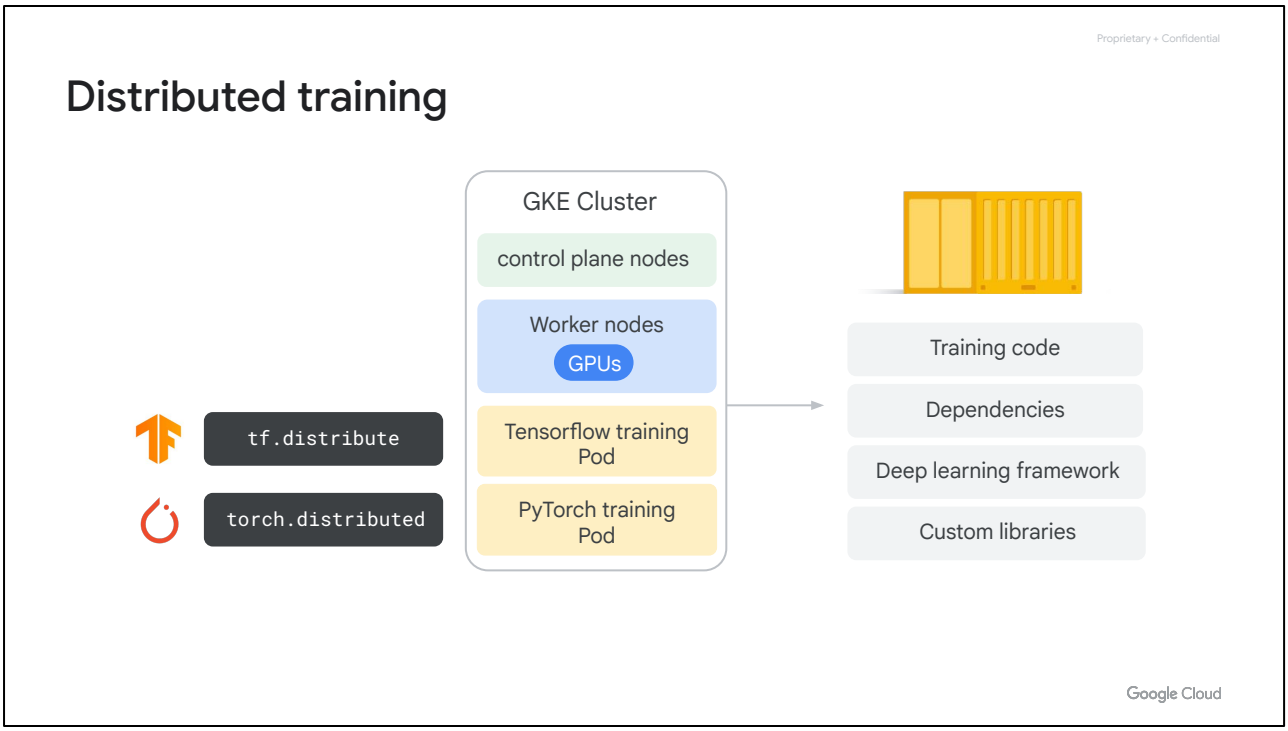
Next, let's examine the GKE setup and implementation details.

First you must provision a GKE cluster and include node pools that are equipped with GPUs for acceleration.

Multiple node pools can be configured to accommodate different training needs. You can also experiment with different hardware configurations.

Kubernetes operators like the Kubeflow Training Operator are used to manage training jobs, or you can develop custom controllers.

If your training data is not accessible via a cloud storage API, you may need to set up persistent volumes, although direct access from Cloud Storage is generally more efficient.



Next, containerize your training application. This involves packaging training code along with all the necessary dependencies, including the chosen deep learning framework (TensorFlow or PyTorch) and any custom libraries.

Distributed training uses TensorFlow's `tf.distribute` API or PyTorch's `torch.distributed` package.

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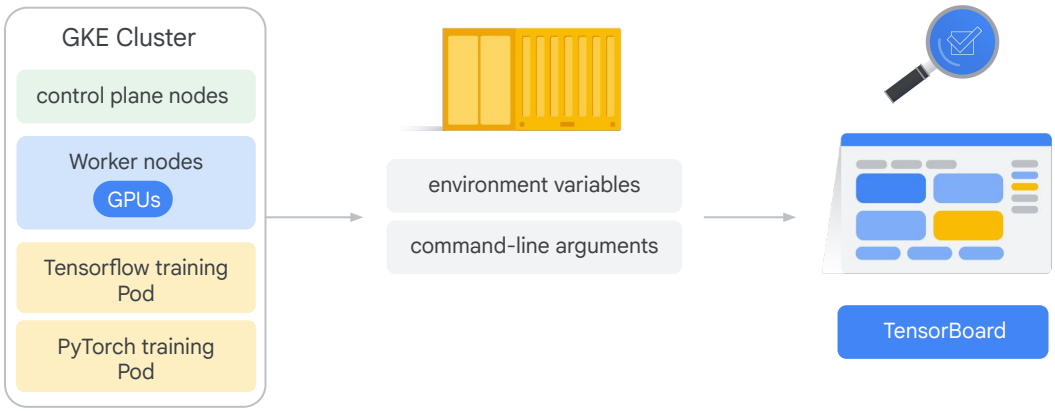
Distributed training



Google Cloud

Configure environment variables and command-line arguments to pass training parameters to your application. To monitor training progress, integrate tools like TensorBoard or other monitoring solutions.

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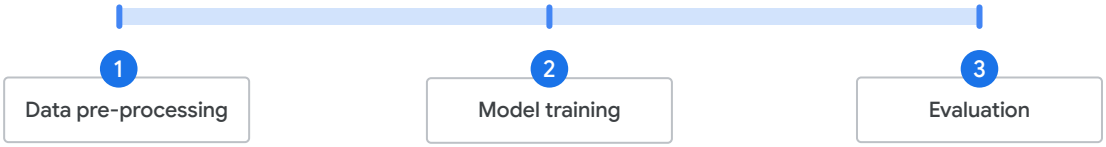
Orchestration tools like Kubeflow pipelines automate the training workflow

Define your training pipeline as a sequence of steps.



Google Cloud

Workflow orchestration tools like Kubeflow Pipelines automate the training workflow.

To automate the execution of the pipeline, define your training pipeline as a sequence of steps, like data pre-processing, model training, and evaluation. This ensures consistency and reproducibility.

For CI/CD, integrate with Cloud Build to automate training application builds and deployments.

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Benefits of using GKE for AI training

Scalability

Easily scale training jobs based on demand.

Flexibility

Support for custom training environments and dependencies.

Control

Fine-grained control over resource allocation and hardware.

Google Cloud

Using GKE for model training has numerous benefits. It offers excellent scalability, allowing you to easily adjust resources based on your workload demands.

GKE provides the flexibility to support custom training environments and dependencies for full control over your training setup.

It enables fine-grained control over resource allocation and hardware choices for efficient resource utilization and cost optimization.

Finally, GKE integrates seamlessly with other Google Cloud services, such as Cloud Storage, Vertex AI, and Cloud Build, creating a comprehensive and powerful AI development platform.

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Benefits of using GKE for AI training

Cost optimization

Efficient resource utilization and cost management.

Integration

Seamless integration with other Google Cloud services.

Google Cloud

Using GKE for model training has numerous benefits. It offers excellent scalability, allowing you to easily adjust resources based on your workload demands.

GKE provides the flexibility to support custom training environments and dependencies for full control over your training setup.

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Deploying AI models for production presents a unique set of challenges.

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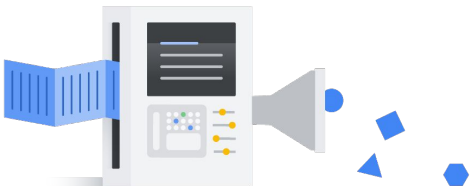
Challenge: deploying AI models



- High volume, low latency inference.
- Scalability and reliability.
- Often require specific dependencies.
- Version management and updates can be complex.

Google Cloud

Many applications demand high-volume, low-latency inference to provide a responsive user experience.

Scalability and reliability are paramount to ensure the service can handle fluctuating workloads without downtime.

Custom serving environments and specific dependencies often must be tailored to your model.

And model version management and updates can be complex.

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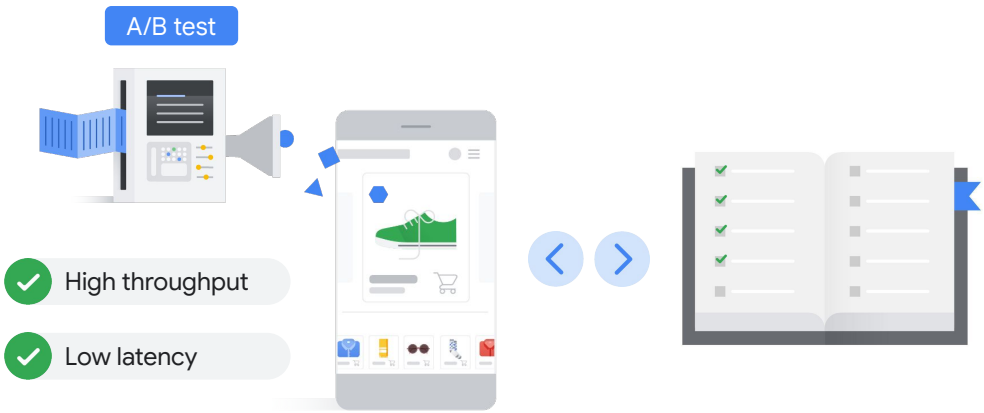
Use Case: real-time recommendation system



Google Cloud

Let's say you're developing a real-time recommendation system. You're serving a machine learning model that provides personalized product recommendations to users on an ecommerce platform.

High throughput and low latency are critical to avoid impacting the user experience.

To optimize performance, you must A/B test different versions of the model.

Finally, seamless integration with other services, like your ecommerce platform's catalog and user database, is essential.

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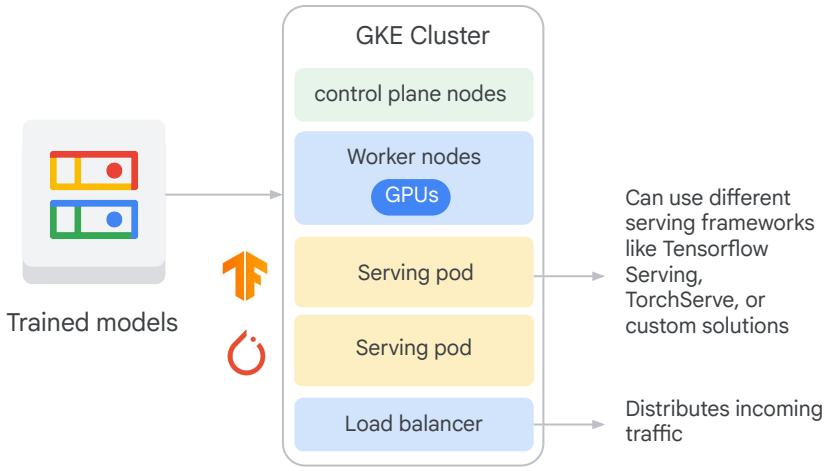
GKE-based serving platform



Google Cloud

The high-level architecture for a GKE-based model serving platform is similar to a training platform.

Trained models are stored in Cloud Storage, and a GKE cluster has control plane and worker nodes.

Serving pods are deployed within the GKE cluster and they can use different serving frameworks like TensorFlow Serving, TorchServe, or even custom serving solutions.

The GKE cluster uses a load balancer to distribute incoming requests, and Vertex AI can be integrated for efficient model registry and management.

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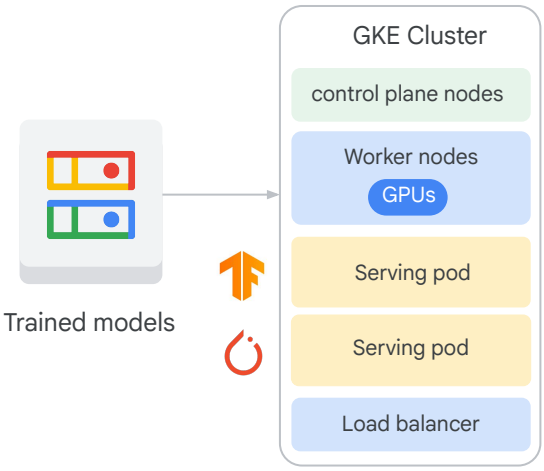
GKE-based serving platform setup

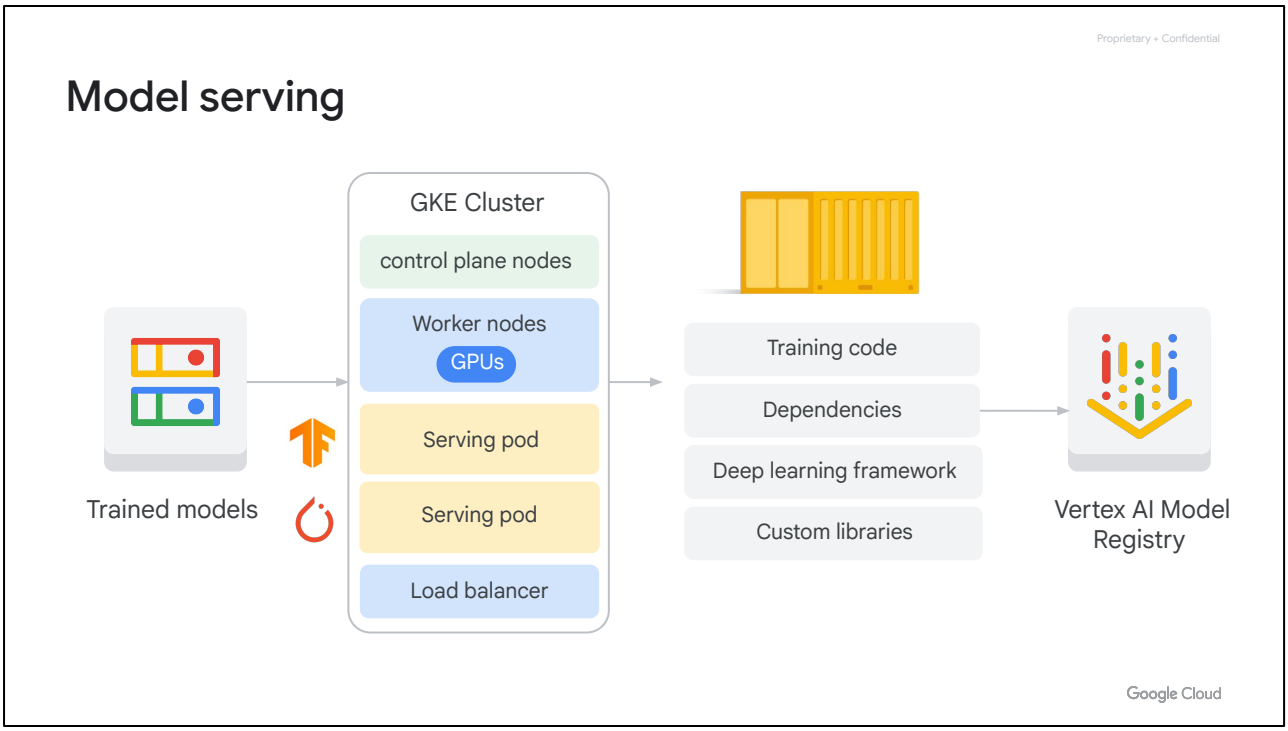


- Kubernetes deployments and services manage scaling and load balancing.
- Ingress or load balancers can be used to serve endpoints to external traffic.

Google Cloud

To implement the platform, you first must provision a GKE cluster specifically optimized for serving. Different node pools might be configured to accommodate different model types or workloads.

Kubernetes deployments and services manage scaling and load balancing within the cluster. Ingress or load balancers can be used to serve endpoints to external traffic.



Next, containerize the model serving application. This includes packaging the model serving code along with all the necessary dependencies, including the chosen serving framework.

You can use frameworks like TensorFlow Serving, TorchServe, or KServe, which are designed for efficient model serving.

If your application requires it, you can implement custom serving logic to handle specific pre-processing or post-processing steps.

This example serving application will load the trained models from Cloud Storage or a model registry like Vertex AI Model Registry.

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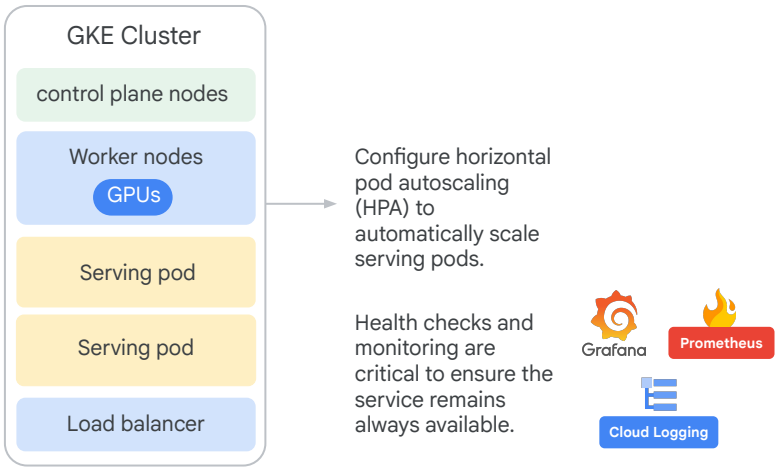
Scaling and monitoring



Google Cloud

To ensure the serving platform can handle varying traffic, configure horizontal pod autoscaling (HPA) to automatically scale the number of serving pods.

Health checks and monitoring are critical to ensure availability. You can use tools like like Prometheus, Grafana, and Cloud Logging to gather metrics like inference latency, throughput, and error rates.

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Benefits of using GKE for model serving

Scalability

Easily scale serving jobs based on demand.

Flexibility

Support for custom serving environments and dependencies.

Control

Fine-grained control over resource allocation and hardware.

Cost optimization

Efficient resource utilization and cost management.

GKE for model serving shares many of the same benefits as GKE for model training, including scalability, flexibility, control, and cost optimization.

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Benefits of using GKE for model serving

Performance

Low latency and high throughput serving.

Availability

High availability and fault tolerance.

It's optimized for high-performance inference, enabling low latency and high throughput serving.

GKE offers high availability and fault tolerance, ensuring your serving platform is reliable.

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Google Cloud

AI and ML workloads have unique infrastructure demands that can be met by running containerized workloads on GKE.

Enterprises aim to build AI/ML platforms that are not only effective but also cost-efficient. GKE is a flexible platform, offering solutions for both managed and unmanaged services.

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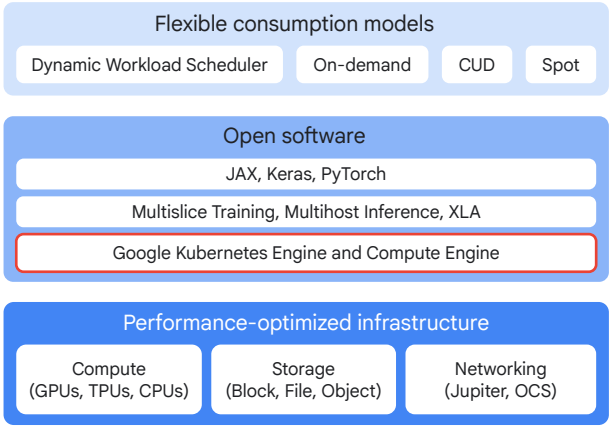
Google AI Hypercomputer architecture with GKE

Efficient scaling optimizes costs, ensuring that you only pay for the infrastructure you need.



Google Cloud

GKE plays a key role in the AI Hypercomputer infrastructure. It is optimized for AI because it can efficiently scale to accommodate increasing complexity and data intensity.

Effective scaling is essential for optimizing costs, ensuring that you only pay for the infrastructure you need.

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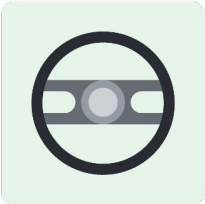
GKE standard and autopilot modes

Standard mode

Custom configurations and flexibility



Autopilot mode

Automatic cluster management

Google Cloud

You can choose from two different modes of operation in GKE: GKE Standard and GKE Autopilot.

GKE Standard provides advanced configuration options and flexibility for hands-on control.

Alternatively, GKE Autopilot offers a hands-free experience, where Google manages cluster provisioning, management, and workload deployment, significantly reducing the operational burden and lowering the barrier to entry.

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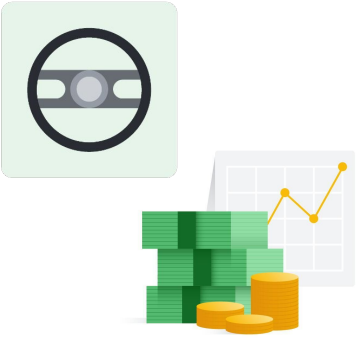
GKE autopilot helps reduce costs



- GKE Autopilot costs are determined by the workload requirements for memory, CPU, and storage.
- There is no cost associated with system pods, operating system overhead, unallocated space, or unscheduled pods.

Google Cloud

GKE Autopilot costs are determined by the workload requirements for memory, CPU, and storage. You don't have to pay for system Pods, operating system overhead, unallocated space, and unscheduled Pods.

This ensures that costs are kept to a minimum when no workloads are running.

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Accelerator optimization is key

- Time sharing allows multiple containers to share a single physical GPU.
- Multi-instance allows larger GPUs to be partitioned into several smaller ones to better fit the needs of your model.
- Multi-process service is an NVIDIA technology that provides an additional way for applications to share a GPU concurrently.

Google Cloud

AI and ML workloads often require hardware accelerators, which can be a significant expense.

There are several GKE innovations available to optimize accelerator usage and reduce operational costs:

- Time sharing allows multiple containers to share a single physical GPU. This can be perfect for workloads like notebooks or low-volume inference.
- Multi-instance allows larger GPUs to be partitioned into several smaller ones to better fit the needs of your model.
- Multi-process service is an NVIDIA technology that provides an additional way for applications to share a GPU concurrently.

By optimizing GPU usage, these innovations collectively reduce GPU spending.

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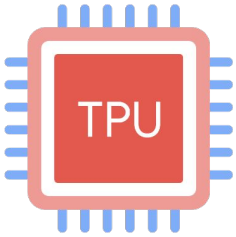
Google TPUs are cost-effective



- A cost efficient, scalable option for a wide range of AI workloads, including training, fine-tuning, and inference.
- Running TPUs on GKE offers excellent price-performance combined with industry-leading scalability.

Google Cloud

Tensor Processing Units, or TPUs, are a cost efficient, scalable option for a wide range of AI workloads, including training, fine-tuning, and inference.

GKE is the first managed Kubernetes offering to support TPUs, providing scalable computing resources.

Running TPUs on GKE offers excellent price-performance combined with industry-leading scalability.

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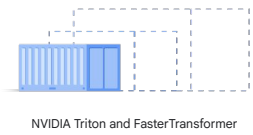
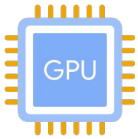
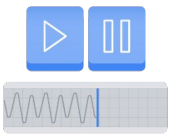
Cost-effective complimentary technologies



Reduce container image size using NVIDIA Triton and FasterTransformer.



Use smaller GPUs for less latency-sensitive workloads.



Using Spot instances for up to 90% cost savings.

And leveraging Cloud Storage FUSE to expose buckets as locally mounted folders with a file cache.

Google Cloud

GKE supports technologies that optimize the use of underlying infrastructure.

Strategies include:

- Reducing container image size using NVIDIA Triton and FasterTransformer.
- Re-evaluating accelerator sizes to use smaller GPUs for less latency-sensitive workloads.
- Using Spot instances for up to 90% cost savings.
- And leveraging Cloud Storage FUSE to expose buckets as locally mounted folders with a file cache.

These approaches help minimize infrastructure costs effectively.

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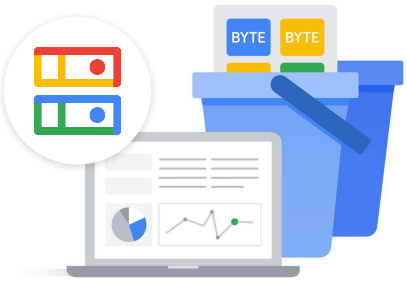
Cloud Storage FUSE



- Direct linking to Cloud Storage for model weights stored in object storage buckets.
- Includes a caching mechanism for frequently read files.
- Does not require pre-hydration operational activities for a pod to download new files in a bucket.
- You can enable parallel downloads, which uses multiple workers to download a model.

Google Cloud

Before wrapping up, let's explore Cloud Storage Fuse further.

Cloud Storage Fuse offers a direct link to Cloud Storage for model weights stored in object storage buckets.

It includes a caching mechanism for frequently read files, which prevents additional downloads from the source bucket and reduces latency.

An advantage of Cloud Storage Fuse is that it doesn't require pre-hydration operational activities for a pod to download new files in a bucket.

However, if you switch buckets, you'll need to restart the pod with an updated Cloud Storage Fuse configuration.

To enhance performance, you can enable parallel downloads, which uses multiple workers to download a model, significantly improving model pull performance.

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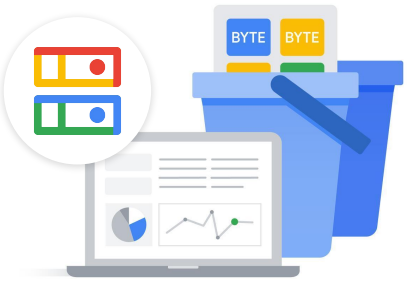
Cloud Storage FUSE benefits



- Improves training time up to 2.3x
- Increases throughput 3.4x
- Performance boost for multi-epoch training workflows
- Improves the speed of input/output operations

Google Cloud

Cloud Storage Fuse increases speed and lowers the cost of AI and ML training by reducing time spent waiting for data.

It improves training time up to 2.3x and increases throughput 3.4x.

This provides a performance boost for multi-epoch training workflows, and it dramatically improves the speed of small, randomly distributed input/output (I/O) operations.